Amy Trent

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EDUCATION

University of British Columbia Sept. 2025 – Apr. 2029

Bachelor of Applied Science | Mechanical and Mechatronics Engineering | 3.9 GPA Kelowna, BC

- **Completed Courses**: Introductory Physics I, Atomic and Molecular Chemistry, Computer Programming I, Differential Calculus, Introductory Physics II, Physical and Organic Chemistry, Microeconomics, Integral Calculus, Engineering Drawing and CAD/CAM, and Linear Algebra.
- **In-Progress Courses**: Statics, Materials Science I, Fundamentals of Sustainable Engineering Design, Calculus III, Instrumentation and Data Analysis, and Thermodynamics.

EXPERIENCE

UBCO Motorsports Oct. 2026 – Present

- Working on the powertrain team to build and optimize an electric race car.
- CAD modeling using Siemens NX.

UBCO Marine Robotics Oct. 2026 – Present

- Designing mechanical components for an autonomous underwater vehicle as a member of the project's mechanical team.

Math and Chemistry Tutor Sept. 2024 – June 2026

- Tutored different math levels, ranging from Grade 5 fundamentals to Grade 11 Pre-Calculus, including quadratic functions, polynomial factoring, rational expressions, linear and quadratic inequalities, radical operations, and trigonometry.
- Held a drop-in tutoring session for students struggling with Grade 11 Chemistry content, including stoichiometry, Lewis structures, and molarity calculations.
- Guided students with failing averages to receive high test marks, up to 100%.

PROJECTS

AI Litter Drone Jul. 2026 – Present

Raspberry Pi 5, YOLOv8, Google Colab

- Designing a drone that uses a Raspberry Pi to detect litter on the ground, fly to it, pick it up, and dispose of it at a garbage or recycling can according to its classification of litter.

Tree Planter Feb. 2026 – Apr. 2026

SolidWorks, 3D Printing, Raspberry Pi, Sensors

- Led a team of six to create a device that helps tree planters be more efficient with planting, while decreasing the back pain that often results from planting at awkward angles.
- Prototype was a university engineering competition finalist.

1989 Batmobile Jan. 2026 – Mar. 2026

SolidWorks

- Led a team of six to design a complete 1989 Batmobile CAD model, designing all parts, assemblies, and mates.
- Modeled steering wheel assembly, frame parts and sub-assemblies, sheet metal accessories, and wing mechanism.

Dragon Head Sept. 2023 – Jan. 2024

C, Arduino MEGA2560, 3D Printing, Servo, Proximity Sensors, 123D Design

- Designed a 3D printed dragon head with proximity sensors that track where a person is in front of it, causing the dragon's LED eyes to follow the motion, a blue LED to signal a "fire attack", and the ears to move according to its position.

Dog Cookie Dispenser Feb. 2021 – Apr. 2021

C, Arduino Nano, 3D Printing, 123D Design

- Designed a 3D printed dog cookie dispenser that played two similar sounds to test whether my dog could recognize which one signified an incoming cookie.

SKILLS

CAD and Fabrication : SolidWorks, Siemens NX, 3D Printing, Bambu Studio.
Hardware : Arduino, Raspberry Pi.
Machine Learning : YOLOv8, Google Colab.
Programming : C, Java, MATLAB, R (RStudio), TypeScript.
Tools : Microsoft Excel, Word, PowerPoint.
Web Development : React, Next.js.

MISCELLANEOUS

Engineering Instagram | 1,500,000+ views
Software Engineering Community Leader | 3,000+ members
Engineering Website and Blog | 2000+ lines of code